Shipment Route Recommendation System

A practical route comparison for the Aiqon take-home assignment

The decision we are trying to support

A shipment can often travel by several routes. The cheapest option may take longer, while a faster route may cost more or be less reliable. This project compares the alternatives supplied for a shipment and recommends one based on the priorities selected by the user. It also explains the choice in everyday terms, so a planner can see what was gained and what was given up.

The sample data

The example dataset contains 30 fictional shipments, each with four route options. A route includes its hubs, estimated cost in US dollars, transit time, estimated delay probability, and a separate context risk index. The data is generated with a fixed seed, so the same sample can be recreated. The route options were deliberately varied to create realistic-looking trade-offs; the figures are illustrative and are not live shipping data or calibrated predictions.

How a recommendation is made

First, the system removes routes that break any limits the user entered, such as a maximum price or delivery time. If every option breaks a limit, it reports that no route qualifies rather than quietly changing the requirement. It then scores the remaining routes on cost, transit time, and delay probability. Each measure is scaled from 0 to 100 within that shipment: the best value receives 0 and the worst receives 100. A weighted average combines the three scores, and the lowest total is recommended.

Profile	Cost	Transit time	Delay risk
Balanced	40%	35%	25%
Cost-sensitive	65%	20%	15%
Time-sensitive	20%	55%	25%

Users can choose a preset profile or adjust the weights themselves. The separate risk index is shown for context but is not added to the score, because doing so could count the same underlying risk twice. Scores are relative to the options for one shipment, so they should not be compared between different shipments.

Why this is useful

The method makes preferences visible instead of hiding them in an unexplained choice. A cost-focused planner can see how the result changes when speed matters more; a hard limit can rule out a route that is unacceptable regardless of its score. The ranked alternatives and short explanation help a person review the recommendation and decide whether the trade-off makes sense. This is useful for comparing known options, but it does not create routes or replace a planner.

Using the project and understanding its limits

Try it from the command line

From the project folder, run **python3 main.py recommend SHP-001**. The result lists the recommended route, its score, and the ranked alternatives. To give speed more importance, use **--profile time_sensitive**. To require a cost below $5,000 and transit under 30 days, add **--max-cost 5000 --max-days 30**. The command **python3 main.py list** shows the sample shipment IDs, and **python3 main.py evaluate** summarizes the example dataset.

There is also an optional browser interface. Run **python3 ui.py** and open **http://127.0.0.1:8000**. It lets the user select a shipment, adjust the three weights, set limits, and compare routes. It runs locally and uses the same scoring code as the command-line version.

One example

For SHP-001, the balanced profile recommends route SHP-001-R4. In the sample it costs $4,880, takes 28 days, and has a 6% estimated delay probability. The runner-up, SHP-001-R1, costs $4,290, takes 37 days, and has an 11% estimated delay probability. So R4 costs $590 more, but is nine days faster and has a 5.6 percentage-point lower estimated delay probability. Someone who values the lowest possible price above speed may prefer R1; the recommendation reflects the balanced weights, not a universal answer.

What the sample evaluation shows

The evaluation applies each preset to all 30 shipments. Every profile found a route within the available options for all 30. The table shows how often the selected route was also cheapest, fastest, or lowest risk. The different profiles favor different criteria, as intended. These results only describe behavior on the constructed sample; they do not measure accuracy on real shipments.

Profile	Mean selected score	Cheapest pick	Fastest pick	Lowest-risk pick
Balanced	41.4 / 100	3 / 30	6 / 30	11 / 30
Cost-sensitive	32.4 / 100	29 / 30	0 / 30	1 / 30
Time-sensitive	32.8 / 100	0 / 30	20 / 30	9 / 30

Limits and next steps

The score depends on which route options are present: adding or removing an option can change the normalized scores. Delay probabilities are treated as if they are known, even though real estimates can be uncertain. The sample also leaves out factors such as capacity, emissions, port schedules, handling costs, and disruption events. Before using a system like this operationally, the data and delay estimates would need to be checked against real outcomes, and the weights and limits agreed with logistics planners.

Project files: README.md for setup instructions; main.py for the command line; ui.py for the optional browser interface; route_recommender.py for scoring; data/routes.csv and data/profiles.json for the sample inputs.